Dmiqwi

Multi-channel AI video generator — tools & pipeline overview

A local Flask app that turns a title and a channel profile into a finished, narrated video: script → voiceover → b-roll/host footage → animated overlays → rendered MP4 — running entirely on this machine except for the external APIs listed below.

1. Pipeline Stages

Every job runs the same ordered pipeline (pipeline/orchestrator.py), persisting progress after each stage so a crash or restart resumes instead of re-paying for already-completed work.

Tool	What it does
1. Outline & hooks	The script model (see §2) drafts a title-driven outline and a set of hook variations before committing to a full script.
2. Script	Full narration is written in the cited author's/persona's voice (script.py), staged per the channel profile — length, tone, and structure all come from the profile JSON.
3. Beat plan	The narration is split into sentence-aware, contiguous "beats" — each beat later gets its own visual (b-roll, image, or animation) and is persisted in plan.json so a resume uses identical beats.
4. Music	Background music is generated via ai33.pro's Suno endpoint, matched to the video's mood/theme.
5. Voiceover	Narration is synthesized in as few large continuous TTS passes as possible (grouped up to ~40k characters per call) rather than one call per beat, then sliced back into per-beat audio+subtitle files — this avoids the pitch/prosody reset that happens at every small-chunk TTS boundary.
6. Host (optional)	If the profile enables an on-camera host, HeyGen generates avatar clips (intro + split-screen inserts) lip-synced to the narration audio.
7. B-roll / images	Each beat is matched to stock footage (Pixabay + Pexels, LLM-curated against the narration text), AI-generated video/images (kie.ai), or a local animation template.
8. Animation overlays	Local, no-API-cost animated graphics (stat cards, comparisons, timelines, etc.) rendered directly with FFmpeg for beats that call for a data/visual aid instead of footage.
9. Render	FFmpeg composites every beat (background + foreground + captions + waveform) in parallel, then concatenates/crossfades the beats into the final MP4 — optionally split into a free "Part 1" and a Patreon-exclusive "Part 2".

Resumability: every stage checks on-disk caches before re-doing work — a crash mid-render never re-pays for a completed script, TTS pass, or fetched b-roll clip; only the missing piece is regenerated.

2. External Service Integrations

Every remote call follows the same async submit-then-poll pattern (pipeline/clients.py) with retry-with-backoff on transient failures.

Tool	What it does
kie.ai — Claude Sonnet 5	Script-model client (pipeline/llm.py) for outlines, hooks, full narration, and the beat plan. A GPT■5.6 key can be swapped in as an alternate script model.
kie.ai — b-roll & images	AI text-to-video b-roll (grok-imagine/text-to-video, exact 6–30s clips) and fallback AI images (Qwen Z-Image) when stock footage doesn't fit a beat.
kie.ai — GPT Image 2	Standalone thumbnail generator (pipeline/thumbnail.py), separate from the video pipeline and billed on its own API key so spend is tracked apart from video generation.
HeyGen	On-camera host avatar clips — intro and split-screen inserts — lip-synced to the narration audio, for profiles with the host enabled.
ai33.pro — text-to-speech	Full-narration voiceover with an accompanying auto-generated SRT transcript, per-language/per-profile voice IDs (ElevenLabs voices via ai33), used for the grouped TTS pass described in §1.
ai33.pro — Suno music	Background music generation matched to the video's theme.
Pixabay + Pexels	Stock b-roll and photo search, cross-referenced so the same clip never repeats within a video; candidates are ranked by an LLM curator against the exact narration they'll cover.

Local, no-API-cost tools

Tool	What it does
FFmpeg render engine	All compositing — backgrounds, foreground overlays, subtitles, waveform visualizations, crossfades/concat — runs locally via FFmpeg. Nothing here costs API credit.
Animation templates	16 local animation styles (checklist, stat, compare, quote, keyword, bar chart, list, warning, timeline, gauge, did-you-know, myth/fact, definition, trio, photo card, chapter) across 2 visual themes, used instead of footage when a beat calls for a graphic.
Captions	SRT/VTT parsing and ASS subtitle generation for burned-in captions, with the timing math to slice a multi-beat transcript into exact per-beat cues.

3. Apple Silicon & Concurrency Tuning

The app auto-detects this machine's hardware (CPU cores via os.cpu_count(), RAM via sysctl hw.memsize) and can size its own concurrency instead of requiring manual tuning.

Tool	What it does
GPU encode toggle	A UI switch enables VideoToolbox hardware H.264 encoding (h264_videotoolbox) instead of software libx264, when available on the machine.
MAX_CONCURRENT_JOBS	How many videos render at once. "auto" sizes this to detected cores/RAM; local rendering is CPU-bound (FFmpeg) while every other pipeline stage is a remote API call, so raising this trades CPU contention for throughput.
BEAT_RENDER_WORKERS	How many beats of a single video are composited in parallel. "auto" sizes to cores, tightened when GPU encode is on since VideoToolbox is a shared hardware block rather than a per-core resource.
Live progress	A progress bar above the job queue shows real-time completion percentage per running job, driven by callbacks fired at each beat-render completion.

4. Channel Profiles

A profile (profiles/*.json) is a complete recipe for one channel's videos: persona/voice, script length and structure, host on/off, per-language voice IDs, b-roll style, and Patreon-split behavior. Switching profiles changes the output channel without touching pipeline code.

Tool	What it does
classic	Joseph Murphy-style narration — no on-camera host, narration written as the cited author teaching directly.
classic-carl / classic-sigmund	Same classic format, voiced as Carl Jung / Sigmund Freud respectively, each with a dedicated ElevenLabs voice ID per language.
grandma-hannah	The original on-camera-host format — HeyGen avatar intro + split-screens over b-roll.
matthias-richter	A dedicated host persona/voice profile.
north-pro	An additional channel profile variant.
test-short	A short, low-cost profile used to smoke-test pipeline changes end-to-end without a long render.

Generated from the current state of the Dmiqwi codebase (pipeline/, profiles/, .env.example).